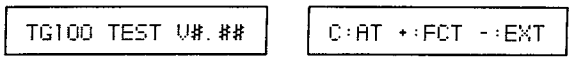
■テストプログラム

A．テストプログラムの起動

[HOST SELECT] スイッチを [MIDI] ポジションにセットし、[MIDI IN] と [MIDI OUT] をMIDIケーブルで接続します。そして、[CURSOR] と [−/NO] と [+/YES] キーを押しながら電源を入れると下の左側の画面が表示され、しばらくすると右の画面が表示されます。



[CURSOR]、[+/YES] または [−/NO] キーを使用してテストモードの選択を行います。

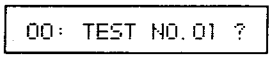
[CURSOR] を押すとオートモードでテストにエントリーされます。

[+/YES]を押すと、テスト11のファクトリーセットを実行した後、自動的にテストモードから抜けてプレイモードになります。

[−/NO] を押すと、テストモードを抜けます。なお、このときはファクトリーセットは実行されません。

B．テストの進め方

オートモードでテストプログラムが起動されると、次の画面が表示されます。この間にイニシャルテストのEEPROM/PSRAMテストが実行されます。



[+/YES] と [−/NO] キーを使用して実行を開始するテスト番号を選択します。

その後、[CURSOR]キーを押すと、設定したテスト番号から順に自動的にテストが実行されます。テスト番号を設定することなしに [CURSOR] キーを押すと、テスト1のLCDテストよりテストが実行されます。

NGと判断された時のテストの進め方

次の各テストにおいてエラーが発生したときは、エラー表示がされてテストが止まります。

テスト3：MIDIテスト

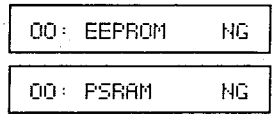
テスト4：TO HOSTテスト

このとき [CURSOR] を押すと、テストはエラーが発生したテストから再開されます。[+/YES] を押すと、テストはエラーが発生したテストの次のテストから再開されます。また [−/NO] を押すと、テストモードを抜けます。なお、テスト2のパネルスイッチテストのときにエラーが発生した場合は、この機能は働きません。

C．イニシャルテスト　EEPROM/PSRAMテスト

EEPROM(IC4)とPSRAM(IC3)のリード/ライトテストが実行されます。テストがOKの場合、“00: TEST NO. 01”が表示され、テスト番号の入力待ちとなります。

判定結果の表示



テストの終了方法

NGと判定した場合、NGを表示して無限ループに入ります。ここから抜けるには、電源を切ります。なお、テストが実行されてもEEPROMの全てのデータは保存されます。

テスト1　LCDテスト

LCDの全ドットがブリンクしていることを、目でみて確認します。また、リアパネルのCONTRASTツマミを動かして、コントラストが連続的に変わることを確認します。

テストの終了方法

[CURSOR] を押すとテストは終了し、次のテストへ進みます。

テスト2　パネルスイッチテスト



[PLAY] から [+/YES] までのパネルスイッチを、次のようなLCDの表示に従ってON/OFFします。



([PART] スイッチのチェックの場合)

押されたスイッチが正常な場合、“ポーッ”と発音して次のスイッチのテストに進みます。表示と違うスイッチが押されると、NGが表示されて発音しません。その後、正しいスイッチが押されると、次のスイッチのテストに進みます。
全てのスイッチが正常であれば、OKが表示されます。

判定結果の表示

```
02: SW          OK
02: PLAY        NG
```

([PLAY] スイッチがNGのとき)

テストの終了方法

[+/YES] までチェックするとOKが表示されてテストは終了し、次のテストへ進みます。
NGが発生した場合、テストはそこでストップし表示しているスイッチが押されるまで無限に待ちます。このルーチンを抜けるには電源を切ります。

テスト 3　MIDIテスト

```
03: MIDI
```

[MIDI IN] と [MIDI OUT] を [MIDI] ケーブルで接続した後、テストを実行します。

判定結果の表示

```
03: MIDI        OK
03: MIDI        NG
```

テストの終了方法

判定結果を表示して終了します。
テストの途中でNGと判断したとき、またはNGと表示したときの処理方法は、“B.テストの進め方”を参照して下さい。

テスト 4　HOST SELECTテスト

```
04: HOST SW  xxx
```

(xxx：現在セットされているスイッチの位置)

このテストは、工場出荷検査用に準備されたテストです。従って、ここでは実行しません。電源を切って、テストを終了して下さい。

テスト 5　TO HOSTテスト

```
05: HOST Tx/Rx
```

このテストは、工場出荷検査用に準備されたテストです。従って、ここでは実行しません。

テストの終了方法

[+/YES]を押すとテストは終了し、次のテストへ進みます。

テスト 6　1kHz OUTPUT L発音

```
06: 1kHz L
```

[LINE OUT(L)]、[LINE OUT(R)]、[PHONES(L/R)] 共にプラグを差し込み、各出力の周波数、出力波形、出力レベルを、周波数カウンタ、オシロスコープ、レベル計(12.47kHzフィルター付き)で観測します。このとき、マスターボリュームは最大とします。

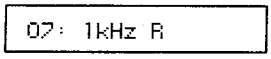
チェック項目

LINE OUT L：1kHz±1.5Hz、サイン波、歪率0.6%以下、+1.0±2dBm(負荷10kオーム)
LINE OUT R：−72dBm以下
PHONES(L)：サイン波、+5.0±2dBm(負荷150オーム)
PHONES(R)：−66dBm以下

テストの終了方法

[CURSOR] キーを押すと発音が終了し、次のテストに進みます。

テスト 7　1kHz OUTPUT R発音



[LINE OUT(L)]、[LINE OUT(R)]、[PHONES(L/R)]共にプラグを差し込み、各出力の周波数、出力波形、出力レベルを、周波数カウンタ、オシロスコープ、レベル計(12.47kHzフィルター付き)で観測します。このとき、マスターボリュームは最大とします。

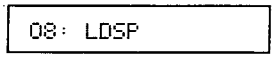
チェック項目

LINE OUT L：−72dBm以下
LINE OUT R：1kHz±1.5Hz、サイン波、歪率0.6%以下、+1.0±2dBm(負荷10kオーム)
PHONES(L)：−66dBm以下
PHONES(R)：サイン波、+5.0±2dBm(負荷150オーム)

テストの終了方法

[CURSOR]キーを押すと発音が終了し、次のテストに進みます。

テスト 8　LDSP/PSRAMテスト



このテストは、テスト11のファクトリーセットを実行した後で行って下さい。

[LINE OUT(L)]、[LINE OUT(R)]、[PHONES(L/R)]共にプラグを差し込み、[LINE OUT(L)]の出力の周波数、出力波形、出力レベルを、周波数カウンタ、オシロスコープ、レベル計(12.47kHzフィルター付き)で観測し、LDSPとPSRAMが正常に動作していることを確認します。このとき、マスターボリュームは最大とします。

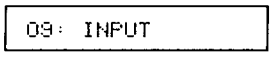
チェック項目

ソフトバージョンV1.06のとき(ROM＝XK731B00)
LINE OUT L：1kHz±1.5Hz、サイン波、−15.0±2dBm(負荷10kオーム)
ソフトバージョンV1.06以上のとき
(ROM＝XK731C00～)
LINE OUT L：1kHz±1.5Hz、−12.0±2dBm(負荷10kオーム)
※ソフトバージョンは、テストプログラム起動時にLCDに表示されます。

テストの終了方法

[CURSOR]キーを押すと発音が終了し、次のテストに進みます。

テスト 9　PEAK LED/INPUT LEVEL CONTROLテスト



[LINE OUT(L)]、[LINE OUT(R)]、[PHONES(L/R)]共にプラグを差し込み、[AUDIO IN]にサイン波を入力したときの各出力の出力波形、出力レベルを、オシロスコープ、レベル計(12.47kHzフィルター付き)で観測します。また、[PEAK LED]が点滅することも確認します。このとき、マスターボリュームは最大とします。

チェック項目

AUDIO IN(L)に−20dBm・1kHzのサイン波を入力したとき(AUDIO IN(R)はグランドに接地)
INPUT LEVEL CONTROL：最小のとき
　LINE OUT L：−24.0±2dBm(負荷10kオーム)
　LINE OUT R：−72dBm以下
INPUT LEVEL CONTROL：最大のとき
　LINE OUT L：−8.5±2dBm(負荷10kオーム)
　LINE OUT R：−68dBm以下

AUDIO IN(R)に−20dBm・1kHzのサイン波を入力したとき(AUDIO IN(L)はグランドに接地)
INPUT LEVEL CONTROL：最小のとき
　LINE OUT L：−72dBm以下
　LINE OUT R：−24.0±2dBm(負荷10kオーム)
INPUT LEVEL CONTROL：最大のとき
　LINE OUT L：−68dBm以下
　LINE OUT R：−8.5±2dBm(負荷10kオーム)

INPUT LEVEL CONTROLを最大とし、AUDIO IN(L)に−12dBm・1kHZのサイン波を入力したとき、PEAK LEDが点灯することを確認します。

テストの終了方法

[CURSOR]キーを押すとテストは終了し、次のテストに進みます。

テスト10　28音発音

```
10: 28 CH OUT 00
```

[LINE OUT L]と[LINE OUT R]にモニター用のアンプとスピーカーを接続した後、テストを実行します。
[LINE OUT L]と[LINE OUT R]より、1チャンネルから28チャンネルの各音が、正常に発音されることを確認します。
発音は約0.5秒、0.1秒間隔で繰り返し行われ、発音中LCDの表示は以下のようになります。

```
10: 28 CH OUT xx
```

(xx：現在発音しているチャンネル番号)

テストの終了方法

[CURSOR]キーを押すとテストは終了し,次のテストへ進みます。

テスト11　ファクトリーセット

```
11: FACT SET
```

各のデータを、ファクトリーセットデータにセットします。
[CURSOR]を押すと、ファクトリーセットされた後テスト12へ進み、テストを抜けてプレイモードとなります。
[−/NO]を押すと、ファクトリーセットされずにテストを抜けてプレイモードとなります。

判定結果の表示

ファクトリーセットされた場合

```
11: FACT SET  OK
```

テストの終了方法

判定結果を表示して終了します。

テスト12　EXIT

このテストに入ると、すぐにテストが実行され、テストモードから抜けてプレイモードとなります。

■ MIDI DATA FORMAT

1. GENERAL

1.1 Application

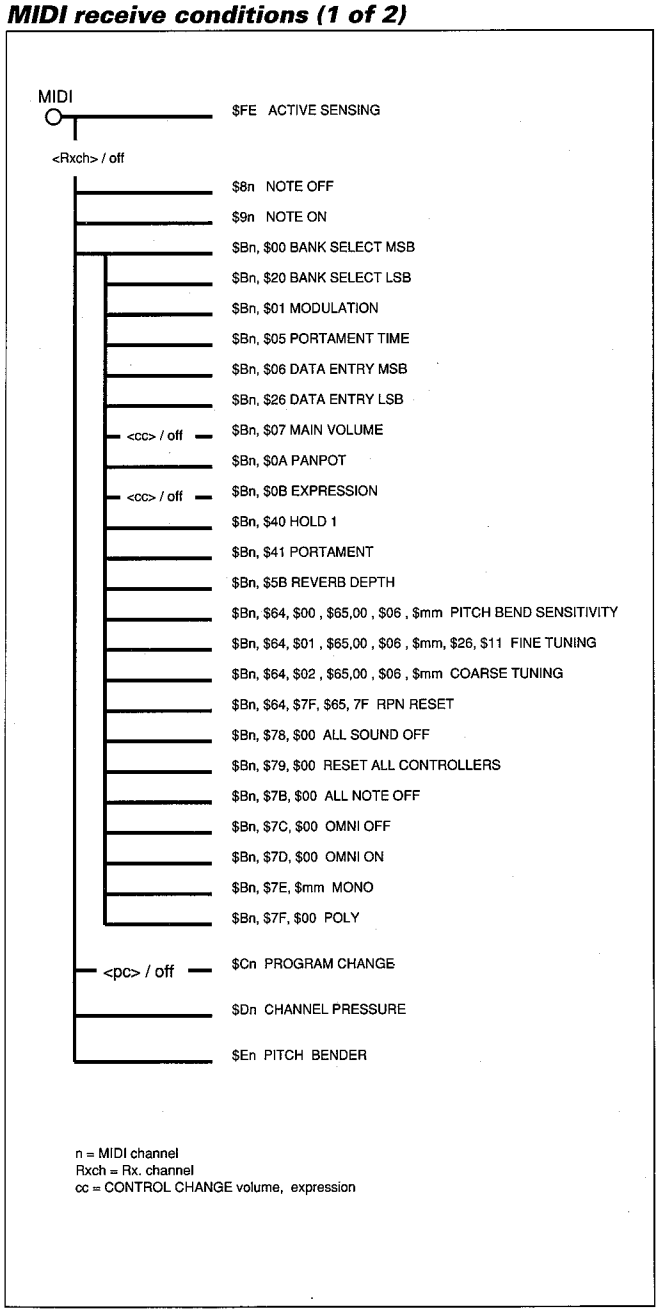
The following MIDI data and specifications apply to the TG100.

1.2 Applied standards

MIDI 1.0 standard.

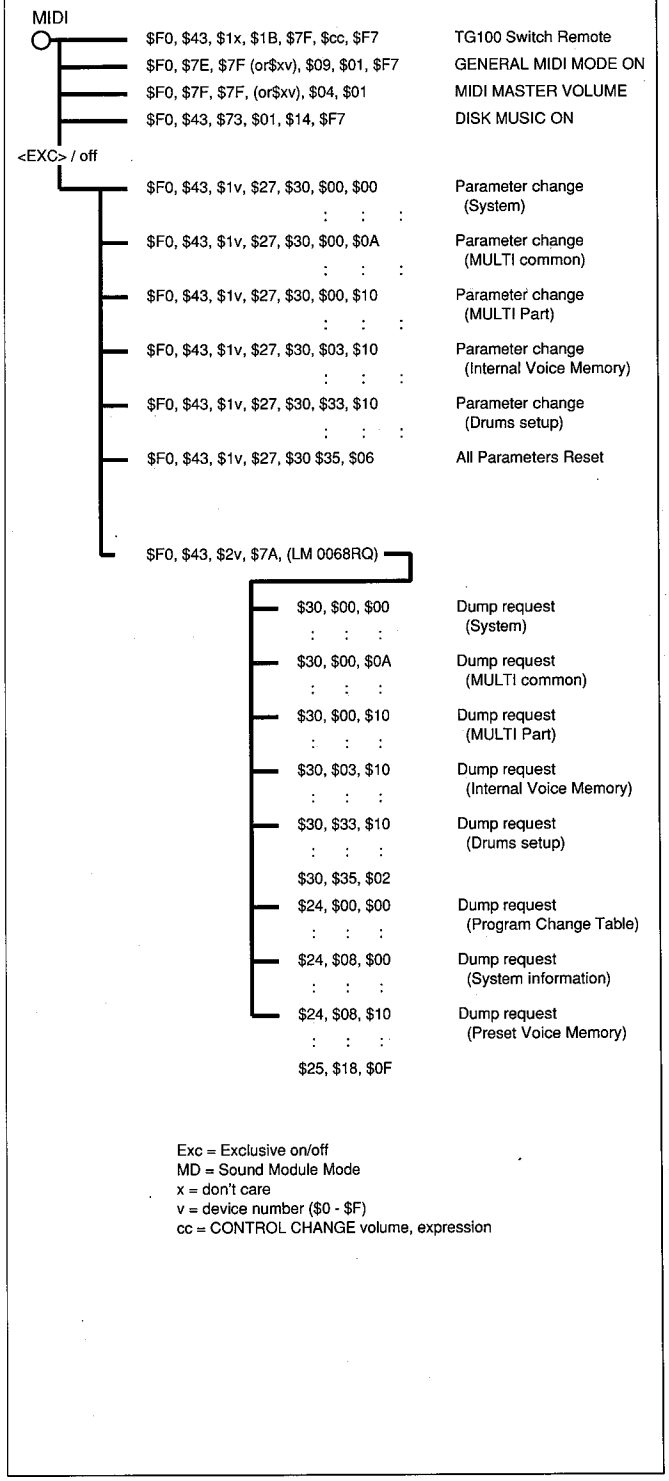
2. MIDI receive & send diagrams

MIDI receive conditions (1 of 2)



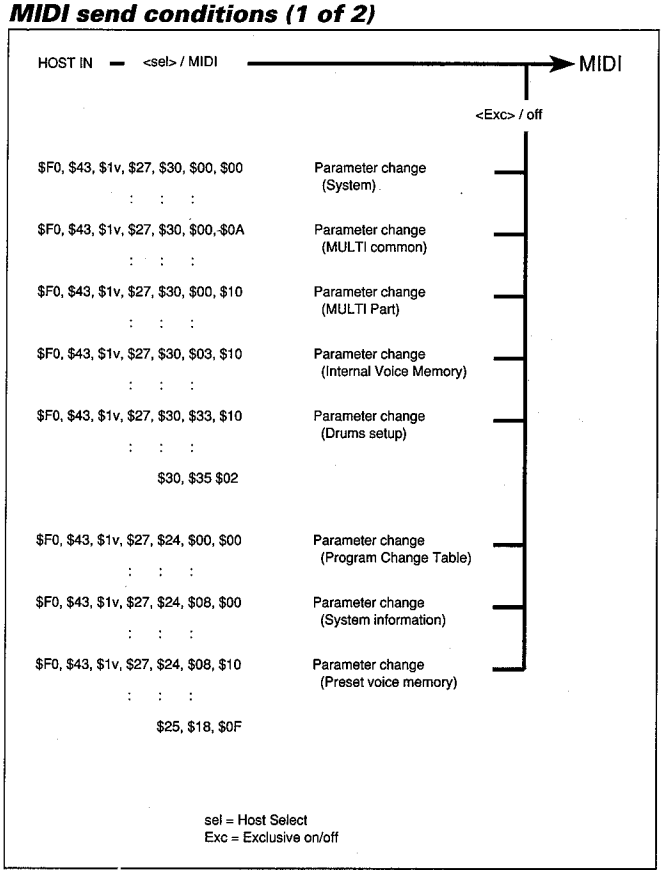
n = MIDI channel
Rxch = Rx. channel
cc = CONTROL CHANGE volume, expression

MIDI receive conditions (2 of 2)



Exc = Exclusive on/off
MD = Sound Module Mode
x = don't care
v = device number ($0 - $F)
cc = CONTROL CHANGE volume, expression

MIDI send conditions (1 of 2)



3. Channel Messages

3.1 Send

Channel Messages not sent.

When the Host Select switch is set to anything other than "MIDI", MIDI data is echoed back as follows.

HOST IN → MIDI OUT

MIDI IN → HOST OUT

3.2 Receive

3.2.1 Note on/off

Note range = C-2...G8

Velocity range 1...127 (only applies to note-on)

3.2.2 Control Change

The following parameters can be controlled using MIDI Control Change messages.

Cntrl No.	Parameter	Data range	
0	Bank Select MSB	0...127	*3
32	Bank Select LSB	0...127	*3
1	Modulation	0...127	
5	Portamento Time	0...127	
6	Data Entry MSB	0...127	*1
38	Data Entry LSB	0...127	*1
7	Main Volume	0...127	
10	Panpot	0...127	*4
11	Expression	0...127	
64	Hold 1	0...127	
65	Portamento	0...127	*2
91	Reverb Depth	0...127	

***1** Used to select parameter value specified by, "3.2.7RPN (Registered Parameter Number)".

***2** Portamento work as follows:

- When a key of a higher pitch than the currently held key is played the pitch sweeps up from a value 100 cents below the key's pitch.
- When a key of a lower pitch than the currently held key is played the pitch sweeps down from a value 100 cents above the key's pitch.

***3** Used to select the unit's voice banks shown below.

MSB	LSB	Bank Name
0...63	0	General MIDI
64...111	0	INTERNAL
112...126	0	DISK ORCHESTRA
127	0	C/M (CM-64)

If a Program Change Message is received immediately after a Bank Select Message has been received, the Program Change number will correspond to the selected voice bank.

***4** When a Part's PANPOT setting is VOICE, the pan position is adjusted relative to the pan position of the elements used by the voice.

When a Part's PANPOT position is not set to VOICE, the elements pan position is ignored and complete adjustment of pan position is possible.

3.2.3 Program Change

You can select one of two Program Change receive modes.

1) off: Ignore Program Changes.
2) on: respond to Program Changers.

In Disk Orchestra mode, if a Program Change number that is not assigned to a voice is received, it is ignored.

In Disk Orchestra, and C/M modes, Program Change numbers are ignored by the drum Part 10.

3.2.4 Pitch Bend

Responds to 14-bit pitch bend data (-8192...+8191).

3.2.5 Channel Pressure

3.2.6 Channel Mode Message

The following Channel Mode messages can be received.

2nd byte	3rd byte	
120	0	All Sound Off
121	0	Reset All Controller
123	0	All Note Off
124	0	Omni Off
125	0	Omni On
126	0...16	Mono
127	0	Poly

3.2.6.1 All Sound Off

Any sound being produced is stopped. However, parts that are receiving Channel Messages such as Note-on, Hold-on etc., will continue.

3.2.6.2 Reset All Controllers

Controllers are set to the following values.

Controller	Reset Value
Pitch Bend	±0 (neutral)
Channel Pressure	0 (off)
Modulation	0 (off)
Expression	127 (maximum)
Hold 1	0 (off)
Portamento	0 (off)
RPN	Not set. Internal data does not change.

3.2.6.3 All Note Off

All Notes currently on will be turned off. However, if Hold 1 is on, sound generation will not stop until Hold 1 stops.

3.2.6.4 Omni Off

Processing is the same as that for "All Note Off".

3.2.6.5 Omni On

Processing is the same as that for "All Note Off".

3.2.6.6 Mono

Processing is the same as that for "All Note Off". If the 3rd byte (Mono value) is between 0...16, the relevant Part is set to Mode 4 (m=1).

3.2.6.7 Poly

Processing is the same as that for "All Note Off". The relevant Part will be set to Mode 3.

3.2.7 RPN (Registered Parameter Number)

Select the control parameter, giving RPN MSB and RPM LSB, then put the parameter value in the Data Entry.

The unit responds to the following RPN.

RPN MSB LSB	Data Entry MSB LSB	
$00 $00	$mm---	Pitch bend sensitivity mm: $00...$18 (0...24 semitones). ---:don't care. A range of 2 octaves can be selected. At switch on range is set to 2 semitones.
$00 $01	mm11	Master fine tuning (mm, 11): ($00, $00)...($40, 400)...($7F, $7F) (-8192 x 100/8192...0...+8191 x 100/8192 cents)
$00 $02	$mm---	Master course tuning mm: $28...$40...$58 (-24...0...+24 semitones) ---: don't care
$7F $7F	--- ---	RPN reset ---; don't care
		RPN Not set. Internal data does not change

4. System Exclusive Message

4.1 Parameter Change

The unit works with the following parameter changes.

1) System Data Parameter Change
2) Multi Common Data parameter change
3) Multi Part Data parameter change
4) Internal Voice Memory parameter change
5) Drums Setup Data parameter change
6) Preset Voice Memory parameter change
7) Program Change Table parameter change
8) System Information
9) All Parameter Reset
10) TG100 Switch remote
11) General MIDI Mode On
12) MIDI Master Volume
13) Disk Music On

Parameter change transmission is switched off only when Exclusive is set to off.

The parameter change format is as follows

```
11110000    F0       = Exclusive status
01000011    43       = YAMAHA ID
0001nnnn    nnnn     = Device Number
00100111    27       = Model ID
0aaaaaaa    aaaaaaa  = Start Address b20 - b14
0aaaaaaa    aaaaaaa  = Start Address b13 - b7
0aaaaaaa    aaaaaaa  = Start Address b6 - b0
0ddddddd    ddddddd  = Data
   |           |
   |           |
0ccccccc    ccccccc  = Check-sum
11110111    F7       = End of exclusive
```

Data is correctly processed if the Dump request receive address corresponds with the Start Address and the Dump Request's byte count is correct.

For the Start Address and byte count, see the tables .

The sending device must add the header to each parameter. For example, when sending System and Multi parameters with only one header, the receiver can only identify System parameters. Therefore, the sending device must not skip these characters.

- System
- Multi
- Internal voice
- Drums Part
- All parameters reset

Do not send more than 256 bytes in one transmission.

If you have a Dump request of more than 256 bytes, split it into sections, then transmit them at 20ms intervals.

The check sum's lowest 7-bit value is zero after adding the Start Address, Data and check sum.

While sending the data, the "HOST" in echo back does not function

4.1.1 System Data Parameter Change

Refer to "Parameter base address" and "MIDI Parameter Change table (System)".

4.1.2 Multi Common Data parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (MULTI common)".

4.1.3 Multi Part Data parameter change

The actual address value = the start address + the offset address.

Refer to "Parameter base address" and "MIDI Parameter Change table (MULTI Part)".

4.1.4 Drums Setup Data parameter change

The actual address value = the address at the top of each block + the offset address.

If a different Drum kit is selected, the Drum Setup parameters are initialized.

Refer to "Parameter base address" and "MIDI Parameter Change table (Drums Setup)".

4.1.5 Internal Voice Memory parameter change

The actual address value = the address at the top of each block + the offset address.

Refer to "Parameter base address" and "MIDI Parameter Change table (VOICE Memory)".

4.1.6 Preset Voice Memory parameter change

This data can be sent, but it is ignored if received.

The actual address value = the address at the top of each block + the offset address.

Refer to "Parameter base address" and "MIDI Parameter Change table (VOICE Memory)".

4.1.7 Program Change Table parameter change

This data can be sent, but it is ignored if received.

Refer to "Parameter base address" and "MIDI Parameter Change table (Program change table)".

4.1.8 System Information parameter change

This data can be sent, but it is ignored if received.

Refer to "Parameter base address" and "MIDI Parameter Change table (System information)".

4.1.9 All Parameters Reset

```
11110000    F0       = Exclusive status
01000011    43       = YAMAHA ID
0001nnnn    nnnn     = Device Number
00100111    27       = Model ID
00110000    30       = Start Address b20 - b14
00111001    35       = Start Address b13 - b7
00000100    06       = Start Address b6 - b0
00000000    00       = Data
00010011    15       = Check-sum
11110111    F7       = End of exclusive
```

Reset the system. All internal parameters are reset to the default (factory) settings.

4.1.10 TG100 Switch remote

```
11110000    F0       = Exclusive status
01000011    43       = YAMAHA ID
0001xxxx    xxxx     = don't care
00011011    1B       = Switch remote ID
01111111    7F       = Switch remote sub ID
0ddddddd    ddddddd  = Data
11110111    F7       = End of exclusive
```

The LCD screen will show the same as when the power is turned on.

The following data is received even if the Exclusive is turned off.

Data	Switch
0	PLAY
1	PART
2	EDIT
3	CURSOR
4	–1/NO
5	+1/YES

4.1.11 General MIDI Mode On

```
11110000    F0       = Exclusive status
01111110    7E       = Universal Non-Real time
01111111    7F       = ID of target device
00001001    09       = Sub-ID No.1=General MIDI
                       Message
00000001    01       = Sub-ID No.2=General MIDI On
11110111    F7       = End of exclusive
```

OR;

```
11110000    F0       = Exclusive status
01111110    7E       = Universal Non-Real time
0XXXnnnn    nnnn     = Device Number, XXX=don't care
00001001    09       = Sub-ID No.1=General MIDI
                       Message
00000001    01       = Sub-ID No.2=General MIDI On
11110111    F7       = End of exclusive
```

The Sound Module mode changes to General MIDI mode when the ON data is received.

The above data is received even if the Exclusive is set to off.

4.1.12 MIDI Master Volume

```
11110000    F0       = Exclusive status
01111111    7F       = Universal Real time
01111111    7F       = ID of target device
00000100    04       = Sub-ID No.1=Device control
                       Message
00000001    01       = Sub-ID No.2=Master Volume
01111111    11       = Volume LSB
0mmmmmmm    mm       = Volume MSB
11110111    F7       = End of exclusive
```

OR;

```
11110000    F0       = Exclusive status
01111111    7F       = Universal Real time
0XXXnnnn    nnnn     = Device Number, XXX=don't care
00000100    04       = Sub-ID No.1=Device Control
                       Message
00000001    01       = Sub-ID No.2=Master Volume
01111111    11       = Volume LSB
0mmmmmmm    mm       = Volume MSB
11110111    F7       = End of exclusive
```

When the "Volume MSB" is received, the Master Volume is set.

The above data is received even if the Exclusive is set to off.

4.1.13 Disk Music On

```
11110000    F0       = Exclusive status
01000011    43       = YAMAHA ID
01110011    73       = Instrument Classified
                       (CLAVINOVA)
00000001    01
00010100    14       = Disk Music On
11110111    F7       = End of exclusive
```

The Sound Module mode changes to Disk Orchestra when the ON data is received.

The above data is received even if the Exclusive is set to off.

4.2 Dump request

The following Dump requests can be carried out.

1) System Data
2) Multi Common Data
3) Internal Voice Memory
4) Preset Voice Memory
5) Program Change Table

Dump request can only be switched off by setting Exclusive to off.

This unit cannot make Dump requests.

Dump requests to this unit should be as follows.

```
11110000    F0        = Exclusive status
01000011    43        = YAMAHA ID
0010nnnn    nnnn      = Device Number
01111010    7A        = Format number
01001100    4C        = "L"
01001100    4D        = "M"
00100000    20        = " "
0010000     20        = " "
00110000    30        = "0"
00110000    30        = "0"
00110110    36        = "6"
00110110    38        = "8"
01010010    52        = "R"
01010001    51        = "Q"
0aaaaaaa    aaaaaaa   = Start Address b20 - b14
0aaaaaaa    aaaaaaa   = Start Address b13 - b7
0aaaaaaa    aaaaaaa   = Start Address b6 - b0
0sssssss    sssssss   = Byte Count b20 - b14
0sssssss    sssssss   = Byte Count b13 - b7
0sssssss    sssssss   = Byte Count b6 - b0
00000000    00
00000000    00
00000000    00
00000000    00
00000000    00
00000000    00
00000000    00
00000000    00
00000000    00
0ccccccc    ccccccc   = Check-sum
11110111    F7        = End of exclusive
```

Data is correctly processed if the Dump request receive address corresponds with the Start Address and the Dump Requests' byte count is correct.

For the Start Address and byte count, see the tables .

The check sum's lowest 7-bit value is zero after adding the Start Address, Data and check sum.

Dump request, the sending device must add the header to each parameter.

For example, when a request for System and Multi parameters is sent with only one header, the unit will send back only the System parameters.

- System
- Multi common
- Multi Part
- Internal voice
- Drums Part
- All parameters reset

4.2.1 System Data parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (System)".

4.2.2 Multi Common Data parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (MULTI common)".

4.2.3 Multi Part Data parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (MULTI Part)"

4.2.4 Drums Setup Data parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (Drums Setup)".

4.2.5 Internal Voice Memory parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (VOICE Memory)".

4.2.6 Preset Voice Memory parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (Program change table)".

4.2.7 Program Change table parameter change

Refer to "Parameter base address".

4.2.8 System Information parameter change

Refer to "Parameter base address" and "MIDI Parameter Change table (System information)".

5. Status FE (active sensing)

A) Sending

Not sent.

B) receiving

After receiving one FE message, if no MIDI data is received for more than 300msec, the unit will activate ALL SOUND OFF, ALL NOTE OFF and RESET ALL CONTROLLERS, as if no FE message had been received.

6. MIDI LCD messages

While exclusive data is being received, there are no messages on the LCD display.

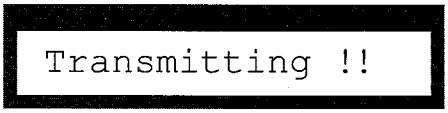
If an error occurs, such as a check sum error, the following message appears; transmission stops, then the previous LCD display appears.



Disconnect the line, and the previously displayed message will appear.

If the device numbers don't match, or the Exclusive is set to off, data is ignored and no message is displayed.

While exclusive data is being transmitted, the following LCD display is shown.



If many MIDI messages are received in too short a time, the following message appears for a few seconds, receiving is stopped, then the previously displayed message appears.



When Program Change messages are received in play mode, the Part, BANK, PC VALUE and VOICE NAME are shown on the LCD display.

1.1 Parameter base address

Parameter change			
Start Address (H)			Description
30	00	00	System
30	00	0A	Multi common
30	00	10	Multi Part 10
30	00	28	Multi Part 1
	:		:
30	02	60	Multi Part 15
30	02	78	Multi Part 16
30	03	10	Internal voice 0
30	03	70	Internal voice 1
	:		:
30	31	70	Internal voice 62
30	32	50	Internal voice 63
30	33	10	Drums Part key No.27
30	33	13	Drums Part key No.28
	:		:
30	35	00	Drums Part key No.107
30	35	03	Drums Part key No.108
30	35	06	All parameters reset
			<Program change table>
24	00	00	General MIDI LEVEL 1
24	02	00	DISK ORCHESTRA
24	04	00	C/M TYPE 1
24	06	00	C/M TYPE 2
24	08	00	System Information
24	08	10	Preset voice 0
24	08	70	Preset voice 1
	:		:
25	16	A0	Preset voice 190
25	17	30	Preset voice 191

Voice Memory

Parameter change		
Offset Address (H)		Description
00	00	Common parameter
00	18	Element 1 parameter
00	3C	Element 2 parameter

The actual address value = the address at the top of each block + the offset address.

1.2 MIDI Parameter Change table (System)

Address (H)			Size (H)	Data (H)	Parameter	Description	Default value (H)
30	00	00	02	1C- E4	MASTER TUNE	-100...+100(cent)	08 00 (80)
30	00	01#				1st b3-0 → b7-4	
						2nd b3-0 → b3-0	
30	00	02	01	28 - 58	TRANSPOSE	-24...+24 semitones	40
30	00	03	01	00 - 10	DEVICE NUMBER	0...15, 16: all	10
30	00	04	01	00 -01	EXCLUSIVE	0: off, 1: on	01
30	00	05	01	00 - 01	PROGRAM CHANGE	0: off, 1: on	01
30	00	06	01	00 - 01	CONTROL CHANGE	0: off, 1: on	
					VOLUME, EXPRESSION		01
30	00	07	01	00 -7F	SOUND MODULE MODE	0: General MIDI 1: Disk Orchestra 2: C/M	00
30	00	08	01	00 -7F	MASTER VOLUME	0...127 (=F0 7F 7F 04 01 xx vv F7)	7F
30	00	09	01	00 - 02	VELOCITY METER	0: off 1: auto 2: on	01
TOTAL SIZE			0A				

REMARKS:

The address marked with "#", cannot be used as the "Start address".

1.3 MIDI Parameter Change table (MULTI common)

Address (H)			Size (H)	Data (H)	Parameter	Description	Default value (H)
30	00	0A	01	00 - 07	REVERB TYPE	0: Hall 1 1: Hall 2 2: Room 1 3: Room 2 4: Plate 1 5: Plate 2 6: Delay 1 7: Delay 2	00
30	00	0B	01	03 - 36	REVERB TIME	3...54	21
30	00	0C	01	18 - 46	REVERB OUTPUT LEVEL	-40...+6dB	3E
TOTAL SIZE			03				

1.4 MIDI Parameter Change table (MULTI Part)

Offset Address (H)		Size (H)	Data (H)	Parameter	Description	Default value (H)
00	00	02	00 - 7F	VOICE BANK	0...63 General MIDI 64...111: INTERNAL 112...126: DISK ORCHESTRA 127: C/M (=Bx 00 vv 20 00)	38
00	01#		00 -7F	PC VALUE	(=Cx vv)	00
00	02	01	00 - 10	RX.CHANNEL	0...15 16: off	n
00	03	01	00 - 01	MONO/POLY MODE	0: Mono (Bx 7E 01) 1: Poly (=Bx 7F 00)	01
00	04	02	1C - E4	DETUNE	-100...+100(cent)	08 00 (80)
00	05#				1st b3-0 $ b7-4 2nd b3-0 $ b3-0 (=Bx 64 01 65 00 06 vv 26 vv)	
00	06	01	28 - 58	NOTE SHIFT	-24...+24 semitones (=Bx 64 02 65 00 06 vv)	40
00	07	01	00 - 7F	VOLUME	0...127 (=Bx 07 vv)	64
00	08	01	00 - 0F	VELOCITY SENSE	0...15	08
00	09	01	00 - 0F	PANPOT	8: voice 9: left : 15: left center 0: center	08

Offset Address (H)	Size (H)	Data (H)	Parameter	Description	Default value (H)
				:	
				7: right	
				(=Bx 0A vv' except voice)	
00 0A	01	00 - 7F	NOTE LIMIT LOW	C-2...G8	00
00 0B	01	00 - 7F	NOTE LIMIT HIGH	C-2...G8	7F
00 0C	01	00 - 08	REVERB SEND DEPTH	0: min	04
				:	
				8: max	
				(=Bx 5B vv')	
00 0D	01	39 - 47	LFO SPEED	-7...+7	40
00 0E	01	31 - 4F	LFO DEPTH	-15...+15	40
00 0F	01	00 - 7F	LFO DELAY	-64...+63	40
00 10	01	39 - 47	EG.ATTACK RATE	-7...+7	40
00 11	01	39 - 47	EG.RELEASE RATE	-7...+7	40
00 12	01	00 - 18	PITCH BEND RANGE	0...24 semitones (=Bx 64 00 65 00 06 vv)	02
00 13	01	00 - 0F	MOD LFO PITCH DEPTH	0...15	0F
00 14	01	00 - 7F	don't care	0...127	00
00 15	01	28 - 58	CAF PITCH CONTROL	-24...+24 (semitone)	40
00 16	01	00 - 0F	CAF LFO PITCH DEPTH	0...15	00
00 17	01	00 - 7F	don't care	0...127	00
TOTAL SIZE	18				

REMARKS:

n: block number (0 - F)

Part 1	n = 1
:	:
Part 9	n = 9
Part 10	n = 0
Part 11	n = A
:	:
Part 16	n = F

x: MIDI channel number (0 - F)

When n = 0 (Drums), the following parameters are ignored.

- PC VALUE
- VOLUME
- PANPOT
- REVERB SEND DEPTH

vv' After conversion, it will be written.

The address marked with "#", cannot be used as the "Start address".

The actual address value = the address at the top of each block + the offset address.

1.5 MIDI Parameter Change table (Drums Setup)

Offset Address (H)	Size (H)	Data (H)	Parameter	Description
00 00	01	00 - 7F	LEVEL	0...127
00 01	01	00 - 0F	PANPOT	9: left
				:
				15: left center
				0: center
				:
				7: right
00 02	01	00 - 08	REVERB DEPTH	0: min
				:
				8: max
TOTAL SIZE	03			

REMARKS:

The actual address value = the address at the top of each block + the offset address.

1.6 MIDI Parameter Change table (VOICE Memory)

1) Common parameter

Offset Address (H)	Size (H)	Data (H)	Parameter	Description	Default value (H)
00 00	01	00 - 01	VOICE MODE	0: 1 element	00
				1: 2 element	
00 01	01	00 - 7F	ELEMENT1 LEVEL	0...127	7F
00 02	01	00 - 7F	ELEMENT2 LEVEL	0...127	7F
00 03	01	20 - 5F	ELEMENT1 DETUNE	-32...+31	40
00 04	01	20 - 5F	ELEMENT2 DETUNE	-32...+31	40
00 05	01	00 - 7F	PORTAMENTO TIME	0...127	01
00 06	01	00 - 0F	MOD LFO PITCH DEPTH	0...15	0F
00 07	01	00 - 7F	don't care	0...127	00
00 08	01	00 - 0F	CAF LFO PITCH DEPTH	0...15	00
00 09	01	00 - 7F	don't care	0...127	00
00 0A	01	00 - 04	ELEMENT 1 PITCH RATE SCALING	0: 100% 1: 50% 2: 20% 3: 10% 4: 5% 5: 0%	00
00 0B	01	00 - 7F	ELEMENT 1 PITCH RATE SCALING CENTER NOTE	0...127 (C-2...G8)	3C
00 0C	01	28 - 58	ELEMENT 1 NOTE SHIFT	-24...+24 semitones	40
00 0D	01	28 - 58	ELEMENT 2 NOTE SHIFT	-24...+24 semitones	40
00 0E	01	00 - 04	ELEMENT 2 PITCH RATE SCALING	0: 100% 1: 50% 2: 20% 3: 10% 4: 5% 5: 0%	00
00 0F	01	00 - 7F	ELEMENT 2 PITCH RATE SCALING CENTER NOTE	0...127 (C-2...G8)	3C
00 10	01	20 - 7F	VC NAME1	ASCII character	
00 11	01	20 - 7F	VC NAME2	ASCII character	
00 12	01	20 - 7F	VC NAME3	ASCII character	
00 13	01	20 - 7F	VC NAME4	ASCII character	
00 14	01	20 - 7F	VC NAME5	ASCII character	
00 15	01	20 - 7F	VC NAME6	ASCII character	
00 16	01	20 - 7F	VC NAME7	ASCII character	
00 17	01	20 - 7F	VC NAME8	ASCII character	
TOTAL SIZE	18				

Remarks:

The actual address value = the address at the top of each block + the offset address.

2) Element parameter

Offset Address (H)	Size (H)	Data (H)	Parameter	Description	Default value (H)
00 00 00 01#	02	00 - 8B	WAVEFORM	0: 139 1st b3-0 → b7-4 2nd b3-0 → b3-0	00 00 (00)
00 02	01	31 - 4F	EG AR	-15...+15	40
00 03	01	31 - 4F	EG RR	-15...+15	40
00 04	01	00 - 7F	LEVEL SCALING BREAK POINT1	C-2...G8	40
00 05	01	00 - 7F	BREAK POINT2	C-2...G8	40
00 06	01	00 - 7F	BREAK POINT3	C-2...G8	40
00 07	01	00 - 7F	BREAK POINT4	C-2...G8	40
00 08 00 09#	02	00 - FF	LEVEL SCALING OFFSET1	-128...+127 1st b3-0 → b7-4 2nd b3-0 → b3-0	08 00 (80)
00 0A 00 0B#	02	00 - FF	OFFSET2	-128...+127 1st b3-0 → b7-4 2nd b3-0 → b3-0	08 00 (80)
00 0C 00 0D#	02	00 - FF	OFFSET3	-128...+127 1st b3-0 → b7-4 2nd b3-0 → b3-0	08 00 (80)
00 0E 00 0F#	02	00 - FF	OFFSET4	-128...+127 1st b3-0 → b7-4 2nd b3-0 → b3-0	08 00 (80)
00 10	01	00 - 0F	PANPOT	9: left : 15: left center 0: center : 7: right	00
00 11	01	00 - 07	LFO SPEED	0...7	04
00 12	01	00 - 7F	LFO DELAY	0...127	00
00 13	01	00 - 7F	don't care	0...127	00
00 14	01	00 - 0F	LFO PITCH MOD DEPTH	0...15	00
00 15	01	00 - 07	LFO AMP MOD DEPTH	0 - 7	00
00 16	01	00 - 01	PITCH LFO WAVE	0: triangle 1: sample & hold	00
00 17	01	00 - 02	P-EG RANGE	0: 1/2 oct 1: 1 oct 2: 2 oct	01
00 18	01	00 - 01	P-EG VELOCITY SWITCH	0: on 1: off	01
00 19	01	00 - 07	P-EG RATE SCALING	0...7	00
00 1A	01	00 - 3F	P-EG R1	0...63	3F
00 1B	01	00 - 3F	P-EG R2	0...63	3F
00 1C	01	00 - 3F	P-EG R3	0...63	3F
00 1D	01	00 - 3F	P-EG RR	0...63	3F
00 1E	01	00 - 7F	P-EG LO	-64...+63	40
00 1F	01	00 - 7F	P-EG L1	-64...+63	40
00 20	01	00 - 7F	P-EG L2	-64...+63	40
00 21	01	00 - 7F	P-EG L3	-64...+63	40
00 22	01	00 - 7F	P-EG RL	-64...+63	40
00 23	01	00 -07	VELOCITY CURVE	0: curve-1 1: curve-2 2: curve-3 3: curve-4 4: curve-5 5: curve-6 6: curve-7 7: curve-8	00
TOTAL SIZE	24				

REMARKS:

The address marked with "#", cannot be used as the "Start address".

In Disk Orchestra mode, the voice velocity curve setting is ignored. It is always set to "curve-8".

The actual address value = the address at the top of each block + the offset address.

1.7 MIDI Parameter Change table (Program change table)

Offset Address (H)	Size (H)	Data (H)	Parameter	Description
00 00 00 01# : : 01 7E 01 7F#	02	00 - FF	SERIAL VOICE# TO PC#1 : : : SERIAL VOICE# TO PC#128	0...191, 255:off voice 0...191, 255:off voice
TOTAL SIZE	100			

REMARKS:

The address marked with "#", cannot be used as the "Start address".

The actual address value = the address at the top of each block + the offset address.

1.8 MIDI Parameter Change table (System information)

Address (H)	size (H)	Data (H)	Parameter	Description
24 08 00	10	23	STRING	ASCII '#'
24 08 01#		30	STRING	ASCII '0'
24 08 02#		30	STRING	ASCII '0'
24 08 03#		36	STRING	ASCII '6'
24 08 04#		38	STRING	ASCII '8'
24 08 05#		20	STRING	ASCII ' '
24 08 06#		20	STRING	ASCII ' '
24 08 0 7#		56	STRING	ASCII 'V'
24 08 08#		45	STRING	ASCII 'E'
24 08 09#		52	STRING	ASCII 'R'
24 08 0A#		3D	STRING	ASCII '='
24 08 0B#		31	STRING	ASCII '1'
24 08 0C#		2E	STRING	ASCII '.'
24 08 0D#		30	STRING	ASCII '0'
24 08 0E#		30	STRING	ASCII '0'
24 08 0F#		20	STRING	ASCII ' '
TOTAL SIZE	10			

REMARKS:

The top address must be the same as the "Start address".

YAMAHA [Tone Generator] Date:14-DEC-1991
Model TG100 MIDI Implementation Chart Version : 1.00

Function ...		Transmitted	Recognized	Remarks
Basic	Default	x	1 - 16	memorized
Channel	Changed	x	1 - 16	
Mode	Default	x	3	
	Messages	x	3,4(m = 1) *2	
	Altered	**********	x	
Note Number		x	0 - 127	
	True voice	**********	0 - 127	
Velocity	Note ON	x	o 9nH,v=1-127	
	Note OFF	x	x	
After Touch	Key's	x	x	
	Ch's	x	o	
Pitch Bender		x	o 0-24 semi	12bit resolution
Control Change	0,32	x	o MSB only	Bank Select
	1	x	o	Modulation Wheel
	5	x	o	Portamento Time
	6,38	x	o	Data Entry
	7	x	o *1	Volume
	10	x	o	Panpot
	11	x	o *1	Expression
	64	x	o	Hold 1
	65	x	o	Portamento
	91	x	o(Reverb)	Effect Depth 1
	100,101	x	o	RPN LSB,MSB
	120	x	o	All Sound Off
	121	x	o	Reset All Cntrls
Prog Change		x	o 0-127 *1	
	True #	**********		
System Exclusive		o *3	o *3	
System Common	Song Pos.	x	x	
	Song Sel.	x	x	
	Tune	x	x	
System Real Time	Clock	x	x	
	Commands	x	x	
Aux Mes- sages	Local ON/OFF	x	x	
	All Notes OFF	x	o(123-127)	
	Active Sense	x	o	
	Reset	x	x	

Notes: *1 ; receive if switch is on.
*2 ; m is always treated as "1" regardless of its value.
*3 ; transmit/receive if exclusive switch is on.

Mode 1 : OMNI ON, POLY Mode 2 : OMNI ON, MONO o : Yes
Mode 3 : OMNI OFF, POLY Mode 4 : OMNI OFF, MONO x : No